

Micrometeoroids and Orbital Debris: Design Considerations

Claire Wadman
15 July 2026
SPCE 5065

Micrometeoroids and Orbital Debris (MMOD): An Overview

- Orbital debris: any man-made objects orbiting the Earth that no longer serve a useful purpose
 - Rocket upper stages, defunct satellites, fragments from satellite collisions or breakups
 - Can reach speeds up to 17,500 mph in low Earth orbit (LEO) (roughly 10x the average speed of a bullet)
- Micrometeoroids: 10 μm -2 mm natural solid objects moving through interplanetary space
- Even very small objects can be destructive when colliding with satellites at extremely high velocities

Evolution of Orbital Debris

- As of 2024, European Space Agency (ESA) reports 36,240 debris objects in orbit

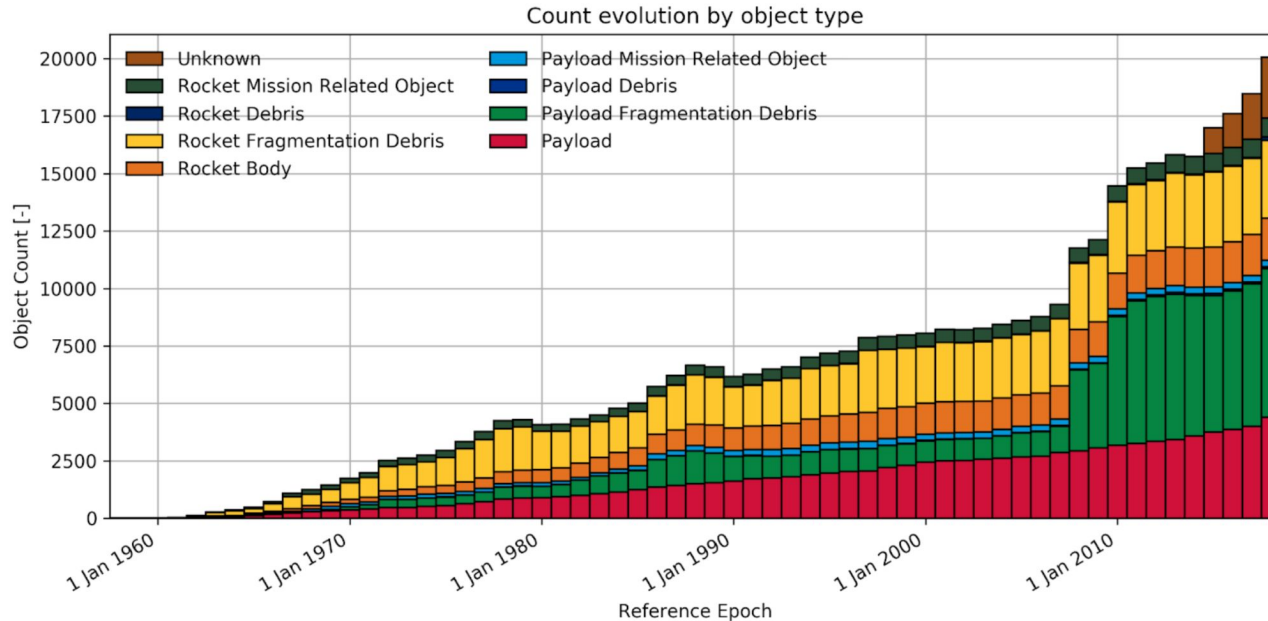


Image courtesy of Clean Orbit Foundation

The Risks

Size	Potential Risk	Potential Impact to Satellite
> 10 cm	Objects are trackable Too large for effective shielding	Catastrophic damage to complete destruction
1-10 cm	Larger objects in range are trackable Still too large for effective shielding	Moderate to catastrophic damage
< 1 cm	Objects are not trackable Effective shielding exists	Minor damage; can still be detrimental to mission

- Results of MMOD impacts range from complete satellite destruction to minor erosions, cracks, and dents
 - However, even minor effects can cause long term damage
 - Result of impact depends on object size and relative velocity between object and satellite

The Risks

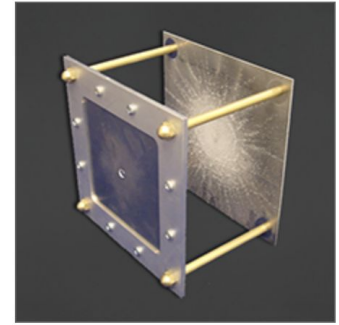


Image courtesy of NASA

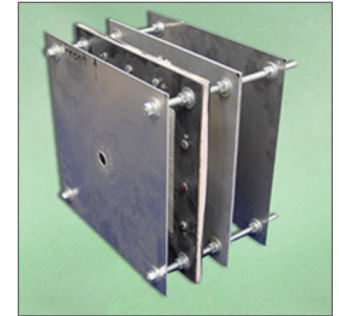
MMOD impact on ISS Zvezda Service Module window

Design Considerations

- Design to withstand impacts by small debris
 - Material selection
 - Shielding (e.g. Whipple Shield)
 - Sacrificial layer(s) in front of spacecraft that absorb shock
- Maintain performance while reducing fragmentation from solar radiation
- Increase quality and reliability of propellants and batteries to prevent accidental explosions
- Improve maneuverability for collision avoidance
- Post-mission disposal
 - Graveyard orbits
 - De-orbit



Whipple Shield



Stuffed Whipple Shield

Thank You!

Sources

- National Space Society. (2023). *Orbital Debris 101*. NSS.org.
<https://nss.org/wp-content/uploads/2023/06/Orbital-Debris-101-An-NSS-Guide-to-the-Orbital-Debris-Issue-and-Its-Management.pdf>
- Asimow, P.D. (2015). *Micrometeoroid*. ScienceDirect.com
<https://www.sciencedirect.com/topics/earth-and-planetary-sciences/micrometeoroid>
- Clean Orbit Foundation. (2024). *Space Debris 101*. CleanOrbitFoundation.org.
<https://cleanorbitfoundation.org/space-debris-101/>
- National Aeronautics and Space Administration (NASA). (2023, October 15). *Micrometeoroids and Orbital Debris*. NASA.gov.
<https://www.nasa.gov/centers-and-facilities/white-sands/micrometeoroids-and-orbital-debris-mmmod/>
- Astromaterials Research and Exploration Science (ARES). *HyperVelocity Impact Technology*. HVIT.JSC.NASA.gov. <https://hvit.jsc.nasa.gov/shield-development/>